

AI Agent Research Compendium — Volume 5: Architecture Blueprints & Production Design Patterns

Compiled from open-source architecture blueprints, production design patterns, and system design documents found on GitHub. Build-ready specifications, not just theory.

Table of Contents

1. [Production Architecture Patterns](#)
2. [Blueprint Codex: 16+ System Blueprints](#)
3. [AI Architecture Blueprints: LangChain/LangGraph Patterns](#)
4. [Enterprise AI Patterns](#)
5. [Additional Architecture Resources](#)

1. Production Architecture Patterns

Agentic Design Patterns: A System-Theoretic Framework (arXiv: 2601.19752)

Five functional subsystems + 12 reusable design patterns:

1. **Perception** — Input processing and environment representation
2. **Reasoning** — Planning, decision-making, inference
3. **Action** — Tool invocation and execution
4. **Memory** — State persistence and retrieval
5. **Communication** — Inter-agent and human-agent interaction

Engineering AI Agents for Clinical Workflows (arXiv: 2602.00751)

Production reference architecture:

- Clean Architecture with bounded contexts
- Event-driven design with CQRS
- Per-agent MLOps lifecycles (training to deployment to monitoring)
- Human-in-the-loop governance with escalation protocols
- Audit trail with immutable action logs

POLARIS: Typed Planning and Governed Execution (arXiv: 2601.11816)

- DAG-based planning with typed plan synthesis
- Rubric-guided selection for agent choice
- Validator-gated execution (every tool call validated)
- Compiled policy guardrails enforced at compile time

MegaFlow: Large-Scale Distributed Orchestration (arXiv: 2601.07526)

- Decouple into Model Service, Agent Service, Environment Service
- Scheduling 10K+ concurrent agent tasks
- Resource isolation and priority queuing
- Distributed tracing and observability

2. Blueprint Codex: 16+ System Blueprints

Source: github.com/ShAuRyA-Noodle/Blueprint-Codex

Blueprint	One-line summary
AURORA	Climate visualization from 40 years of satellite imagery
CHRONOS	Institutional knowledge engine for departing employees
DARWIN	Research validation with simulated experiments
ECHOMIND	Expert reasoning capture via structured interview
HIVEMIND	Collective decision-making as coordinated organism
MERIDIAN	Real-time global sentiment and narrative intelligence

NEXUS	AI software factory: describe product, agents build it
OMNISCIENT	Scientific reasoning engine across research bodies
PHANTOM	Competitive intelligence from weak public signals
PROMETHEUS	Founder toolkit: one idea in, full startup package out
SENTINEL	Planetary crisis prediction via multi-source fusion
SHIELD	Autonomous red-team platform in continuous loop
SUPPLYMIND	Supply chain disruption forecasting
SYMPHONY	Creative AI agent swarm from brief to product

Each blueprint covers: Problem + Architecture + Agent Design + Evaluation + Deployment Phases.

3. AI Architecture Blueprints: LangChain/LangGraph Patterns

Source: github.com/pristley/ai-architecture-blueprints

Foundations (2.5 hours)

- ADR-1.2: Chain abstraction selection (Runnable protocol)
- Working production code examples

Production Patterns (5 hours)

- WP-1.4: Prompt management (versioning, composition, A/B testing)
- WP-1.5: Output parsing (reliability, recovery from failures)
- WP-1.6: Model selection (decision matrix)
- WP-1.7: Observability (LangSmith tracing)

Memory, State & Agents (4 hours)

- WP-2.1: Dual-memory architecture (short-term + long-term)
- WP-2.2: State management and loop detection
- 30+ test suites with production implementations

Multi-Agent Architectures (10 hours)

- ADR-2.1: Choreography (event-driven, emergent)
- ADR-2.2: Orchestration (centralized, deterministic)
- WP-2.6: LangGraph framework (60% less boilerplate)
- WP-2.7: Checkpointing and Human-in-the-Loop

Capstone RAG Patterns (8-10 hours)

- WP-3.1: Naive Baseline (5 failure modes)
- WP-3.2: Advanced Retrieval (Reranking and Filtering)
- WP-3.3: Hierarchical Indexing (100K+ documents)
- WP-3.4: Evaluation and Metrics

4. Enterprise AI Patterns

Source: github.com/SwapnilPopat/enterprise-ai-patterns

Production-ready design patterns for:

- **RAG Architectures:** Naive, advanced, hierarchical, agentic RAG
- **Agentic AI Patterns:** ReAct, Plan-and-Execute, autonomous loops
- **Scalability:** Horizontal agent scaling, load balancing, priority queuing
- **Governance:** Policy enforcement, audit logging, compliance
- **Cost Optimization:** Model routing, token budgeting, caching strategies

Also: github.com/shrikantkashi2510-gif/enterprise-agentic-suite

- Zero-Trust architecture for high-compliance deployments
- Enterprise Context Mesh + AI Ops Agent + Eval Platform

5. Additional Architecture Resources

Key Architecture Papers

Paper	arXiv	Focus
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Agentic AI: Architectures, Taxonomies, Evaluation	2601.12560	Unified agent taxonomy
AI Agent Systems	2601.01743	Components and trade-offs
12 Reusable Design Patterns	2601.19752	System-theoretic patterns
Clinical Workflows Architecture	2602.00751	Production reference
MegaFlow: Distributed Orchestration	2601.07526	Enterprise-scale scheduling
POLARIS: Governed Execution	2601.11816	Governance framework
CaveAgent: Stateful Runtime	2601.01569	Stateful execution architecture
InfiAgent: Infinite-Horizon	2601.03204	Bounded context unbounded tasks
Multi-Agent System Orchestration	2601.13671	MCP + A2A protocol framework

Awesome Lists for Agent Research

Repository	Items	Focus
VoltAgent/awesome-ai-agent-papers	363+	2026 arXiv papers
aloth/awesome-ai-agents	100+	Frameworks, tools, papers
trycua/acu	100+	Computer-use agents
wmmthu/awesome-llm-agent-skills-papers	50+	Agent skills, tool use
kodustech/awesome-ai-code-review	48+	AI code review

Note: No illegally obtained or confidential internal documents were found. All architecture documents found were intentionally open-sourced. The most valuable resources were Blueprint-Codex (16+ complete system specs) and enterprise-ai-patterns repos.